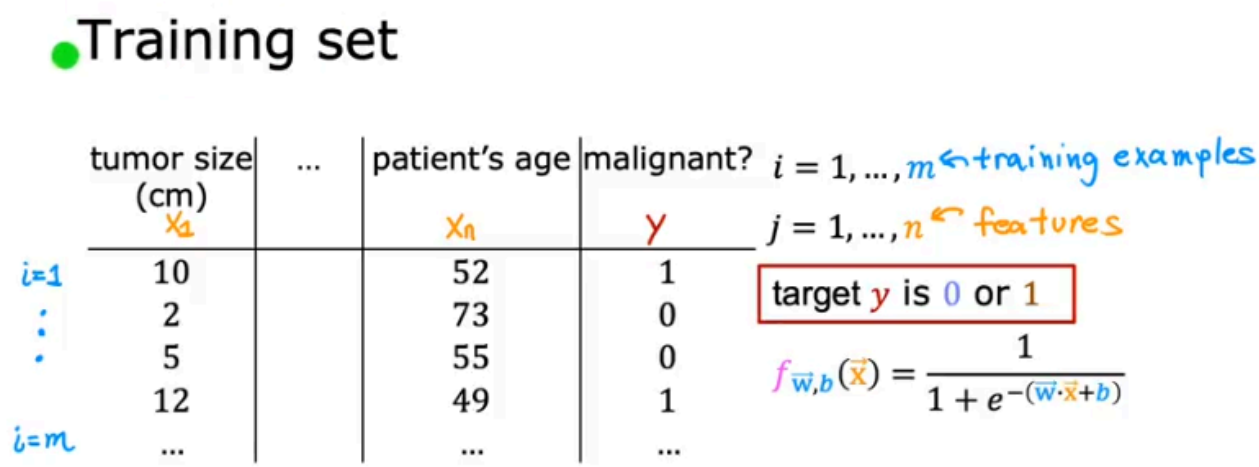
Day 9. Cost function for logistic regression

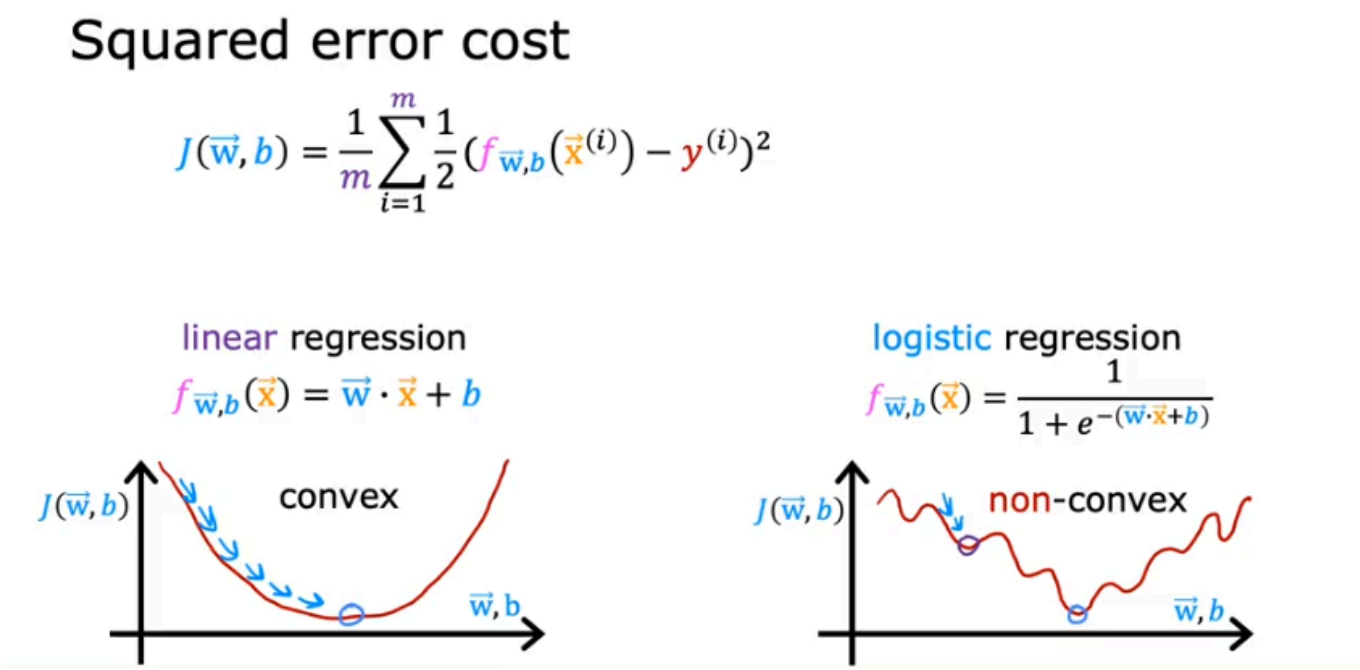
Cost function for logistic regression



RECALL

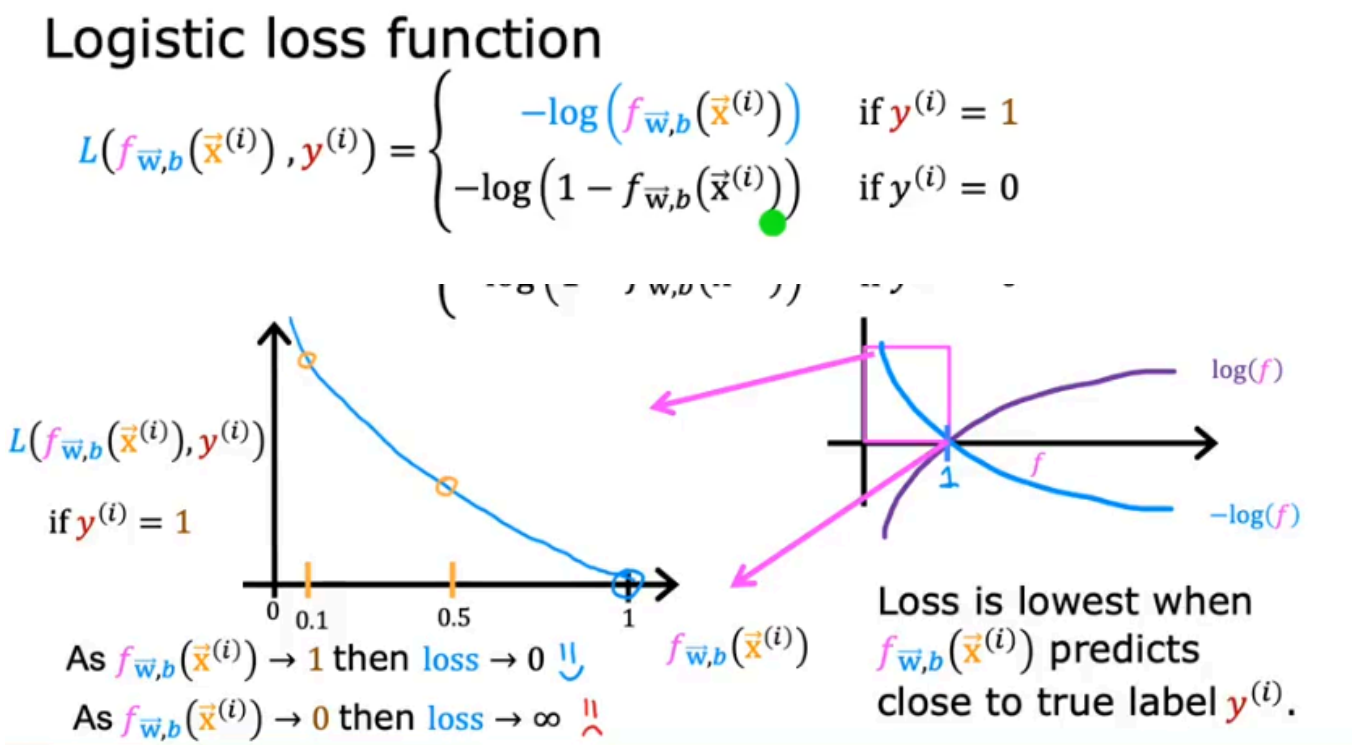
squared error cost in linear regression

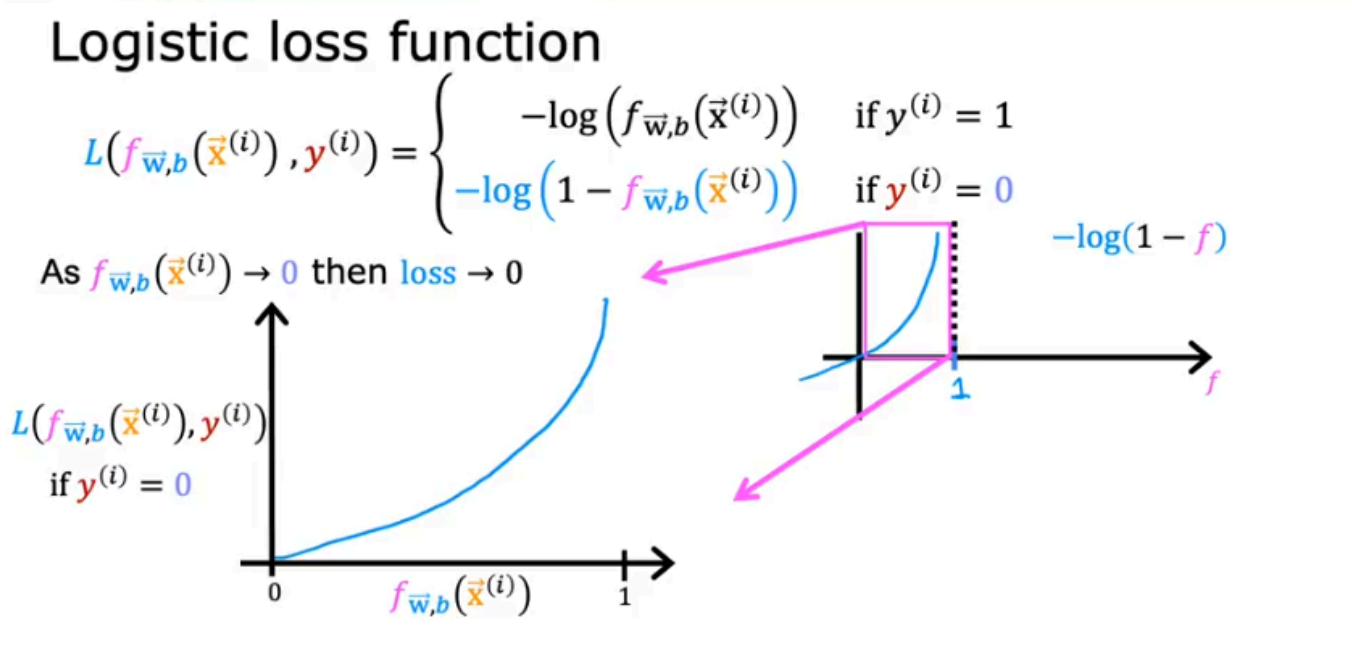
$$J(\vec{w}, b) = \frac{1}{2m}\sum_{i=1}^{m}(f_{\vec{w},b}(\vec{x}^(i)) - y^{(i)})^2$$

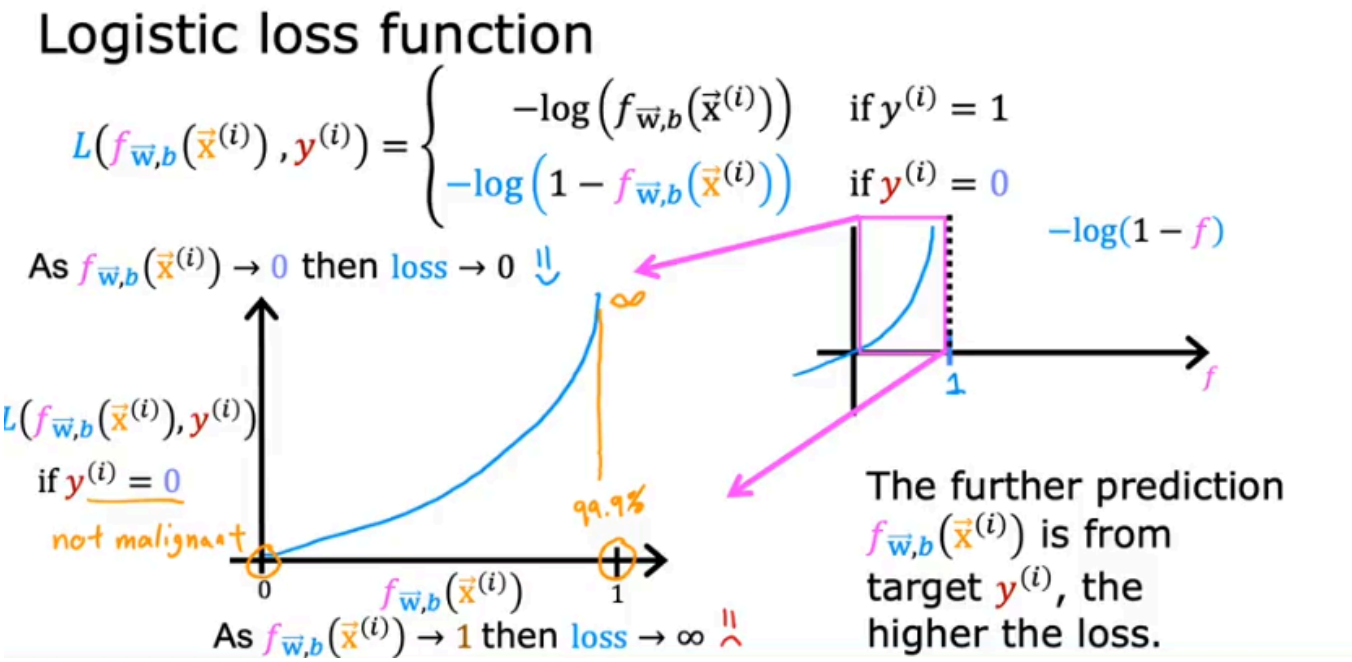


> because the function is non-convex function, it will be difficult for gradient descent to find the right position.

let's call the inside function of the summation $loss function$,by controlling the form of the loss function, we will be able to keep the overall cost function to be a convex function.







so the new cost function will be like

Cost

cost

$$J(\vec{w}, b) = \frac{1}{m}\sum_{i=1}^{m} \underbrace{L(f_{\vec{w},b}(\vec{x}^{(i)}), y^{(i)})}_{\text{loss}}$$

$$= \begin{cases} -\log(f_{\vec{w},b}(\vec{x}^{(i)})) & \text{if } y^{(i)} = 1 \\ -\log(1 - f_{\vec{w},b}(\vec{x}^{(i)})) & \text{if } y^{(i)} = 0 \end{cases}$$

convex
↳ can reach a global minimum

Simplified cost function for logistic regression

$$\boxed{L(f_{\vec{w},b}(\vec{x}^{(i)}), y^{(i)}) = -y^{(i)}\log(f_{\vec{w},b}(\vec{x}^{(i)})) - (1 - y^{(i)})\log(1 - f_{\vec{w},b}(\vec{x}^{(i)}))}$$

just some simple math trick.

cost

$$J(\vec{w}, b) = \frac{1}{m}\sum_{i=1}^{m}[L(f_{\vec{w},b}(\vec{x}^{(i)}), y^{(i)})]$$

$$= -\frac{1}{m}\sum_{i=1}^{m}\left[y^{(i)}\log(f_{\vec{w},b}(\vec{x}^{(i)})) + (1 - y^{(i)})\log(1 - f_{\vec{w},b}(\vec{x}^{(i)}))\right]$$

the reason why we choose this cost function and loss function comes from statistic, so we won't

cover why and how this is happening here, we will cover it later when we learn probability and statistics.